PV102 Day 3 Student Packet

Load Profile and Energy Budget

PV102 | CTS Day 19 | Student Print Packet
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

Today's rule

Watts are moments. Watt-hours are work over time.

Learning targets

- Distinguish watts from watt-hours.
- Calculate daily energy use from power and time.
- Rank loads by necessity, runtime, surge risk, and design impact.
- Explain how load decisions affect PV, battery, inverter, and budget.

Morning quiz

1. A 60 W load runs for 3 hours. How much energy is used?

- A. 20 Wh
- B. 60 Wh
- C. 180 Wh
- D. 3 Wh

2. Watts measure:

- A. Instant power
- B. Total daily energy only
- C. Wire length
- D. Battery age

3. Watt-hours measure:

- A. Energy over time
- B. Compass direction
- C. Module frame size
- D. Fuse color

4. The load that often changes inverter selection is:

- A. Sticky note
- B. Surge or motor load
- C. Pencil
- D. Label maker

5. A good client load interview asks:

- A. What must run, when, and for how long?
- B. What color is the inverter?
- C. How fast can we buy parts?
- D. Which battery looks best?

6. Calculate: 12 V x 5 A = ___ W.

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7. Why should optional loads be separated from essential loads?

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8. Name one load unknown that must be verified before final sizing.

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9. What happens to required battery capacity when runtime doubles?

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10. Write the Day 3 rule in your own words.

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Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Client Load Interview and Energy Budget

Teams interview a mock client, build a daily load table, and recommend one load reduction.

Scenario options

- Tiny shop: lights, laptop, phone charging, modem/router, small fridge, occasional circular saw.
- Remote trail counter: sensor, camera, cellular modem, winter operation, limited service visits.
- Bus conversion: lighting, fridge, fan, water pump, laptop, phone charging, occasional induction cooktop.

Team procedure

- Choose one scenario and list all loads.
- For each load, record watts, hours per day, daily Wh, priority, and unknowns.
- Mark surge or motor loads.
- Calculate total daily Wh and essential daily Wh.
- Recommend one load change that improves system practicality without weakening the client need.
- Prepare a handoff showing the load table and top three unknowns.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which load drove the design hardest today: highest watts, longest runtime, or highest priority? Explain.

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